lora-smd-rak

smd_lora

Table of contents

Sheet 1: Project overview (this page)

Legend

General comment such as function title, configuration, ...

Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

U_breakout
breakout.SchDoc

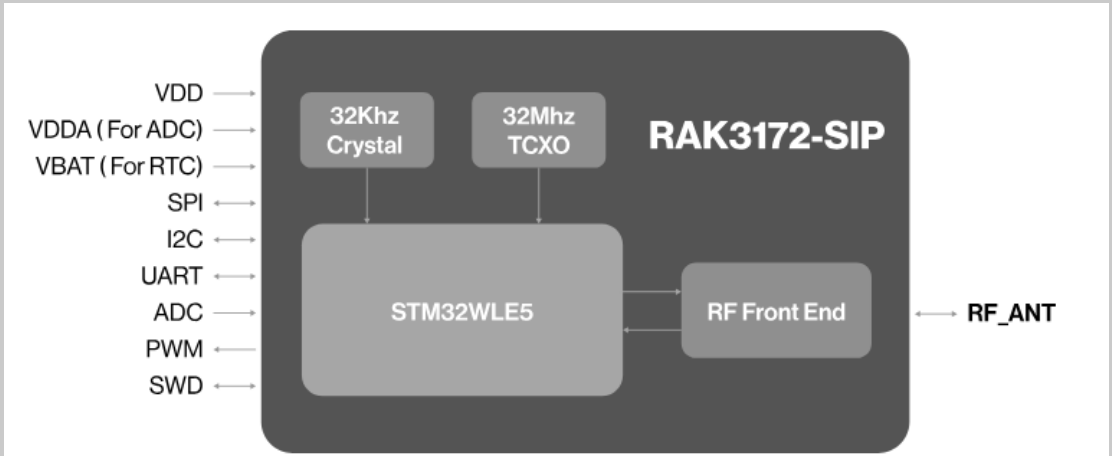
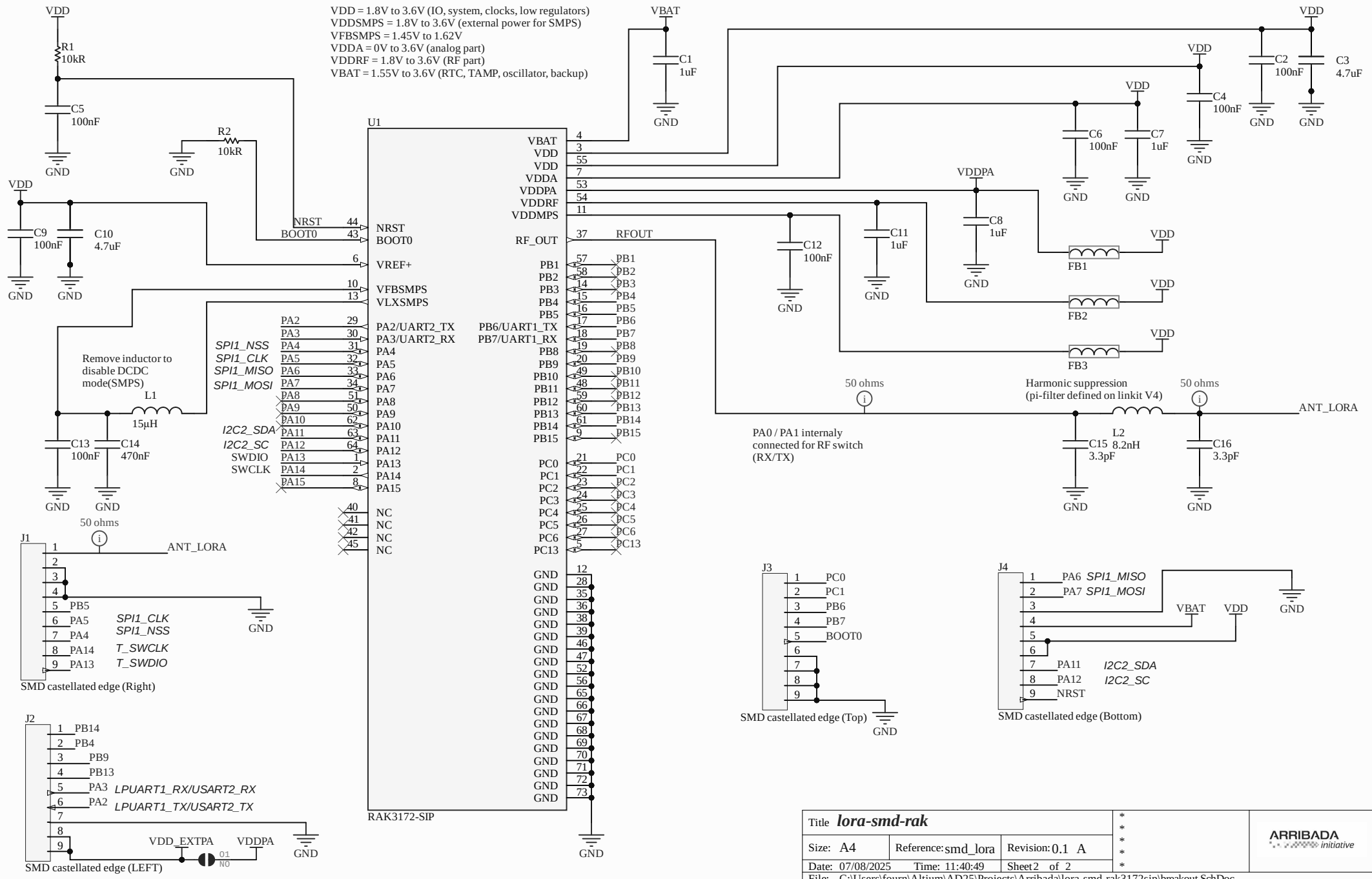


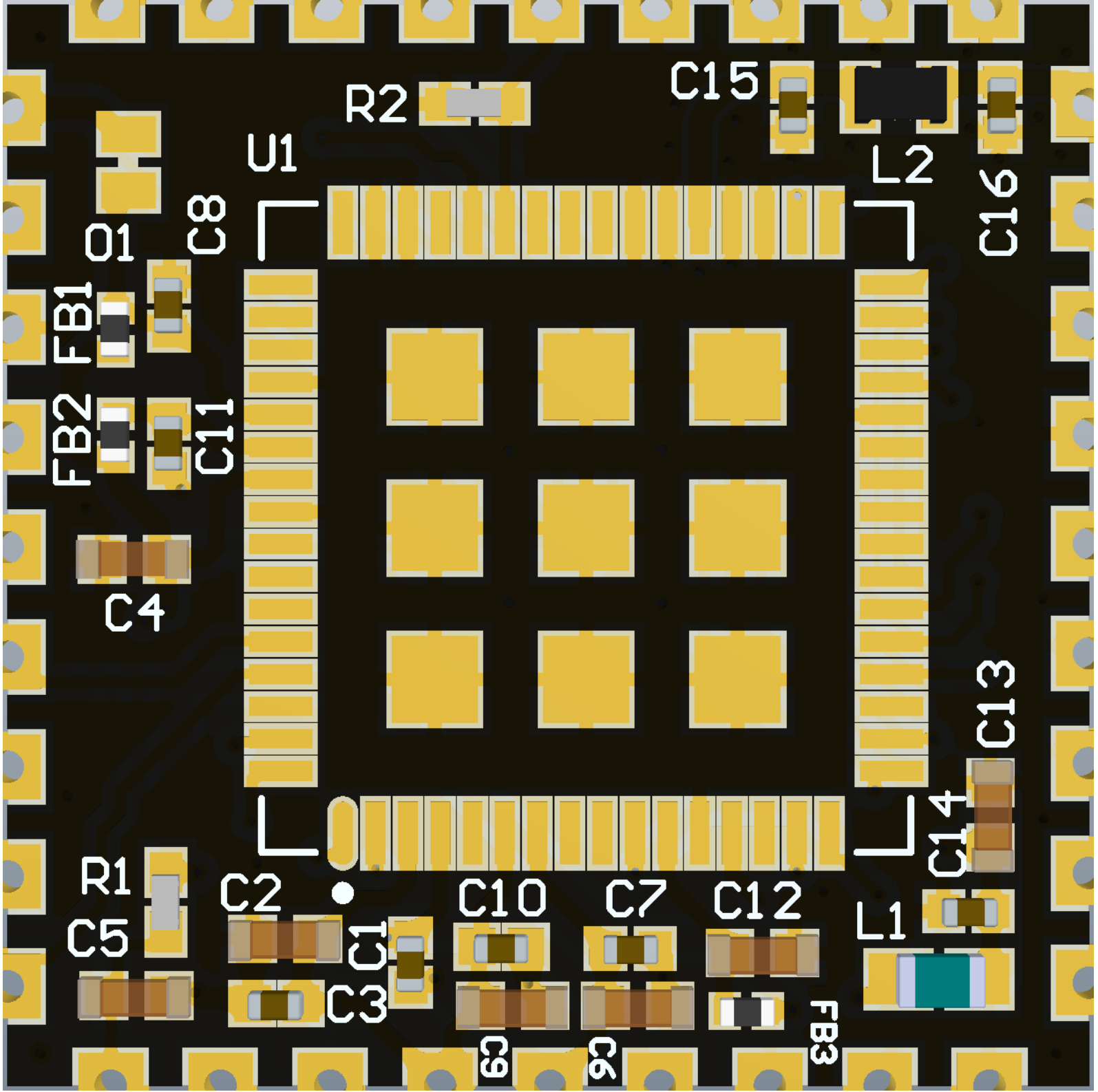
Figure 1: RAK3172-SiP system block diagram

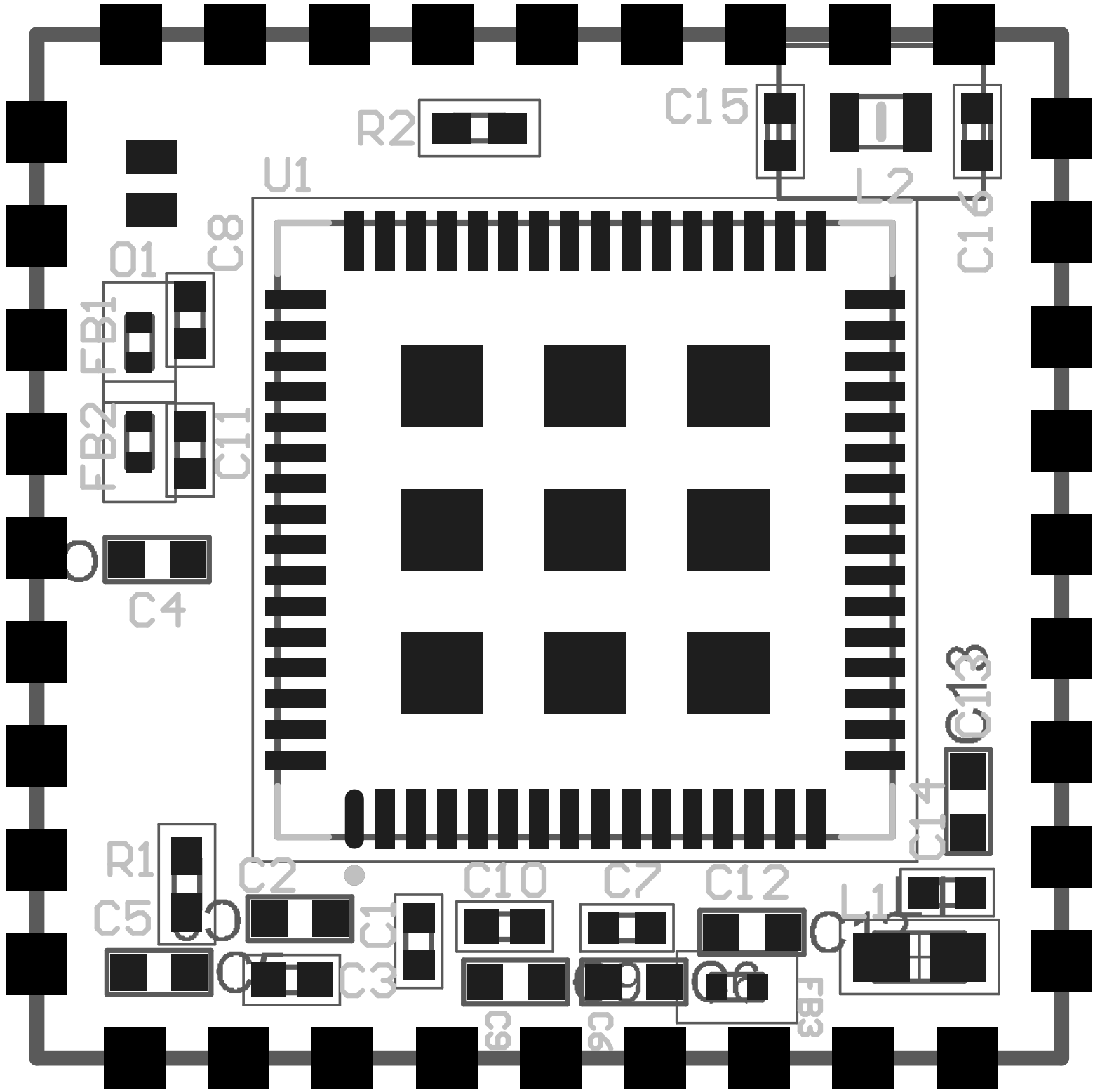
License

Title	***lora-smd-rak***			*Arribada Initiative*
Size: A4	Reference: smd_lora	Revision: 0.1 A		*https://github.com/arribada*
Date: 07/08/2025	Time: 11:40:49	Sheet 1 of 2		*contact@arribada.org*
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\lora-smd-rak3172sip\Project_Overview.SchDoc				*arribada.org*
				UK - Portsmouth

ARRIBADA initiative


VDD = 1.8V to 3.6V (IO, system, clocks, low regulators)
VDDSMPS = 1.8V to 3.6V (external power for SMPS)
VFBSMPS = 1.45V to 1.62V
VDDA = 0V to 3.6V (analog part)
VDDRF = 1.8V to 3.6V (RF part)
VBAT = 1.55V to 3.6V (RTC, TAMP, oscillator, backup)
U1
RAK3172-SIP
Remove inductor to disable DCDC mode(SMPS)
PA0 / PA1 internaly connected for RF switch (RX/TX)
Harmonic suppression (pi-filter defined on linkit V4)
50 ohms
ANT_LORA
SMD castellated edge (Right)
SMD castellated edge (Top)
SMD castellated edge (Bottom)
SMD castellated edge (LEFT)
SPI1_NSS
SPI1_CLK
SPI1_MISO
SPI1_MOSI
I2C2_SDA
I2C2_SC
SWDIO
SWCLK
T_SWCLK
T_SWDIO
LPUART1_RX/USART2_RX
LPUART1_TX/USART2_TX
PB6/UART1_TX
PB7/UART1_RX
VDD_EXTPA
VDDPA
Title lora-smd-rak
Size: A4
Reference: smd_lora
Revision: 0.1 A
Date: 07/08/2025
Time: 11:40:49
Sheet 2 of 2
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\lora-smd-rak3172sip\breakout.SchDoc
ARRIBADA initiative


R2
C15
L2
C16
U1
C8
O1
FB1
FB2
C11
C4
C13
C14
R1
C2
C10
C7
C12
L1
C5
C1
C3
C9
C6
FB3


R2
C15
L2
C16
U1
C8
O1
FB1
FB2
C11
C4
C13
C14
R1
C2
C10
C7
C12
L1
C5
C1
C3
C9
C6
FB3

Line #	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	1uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1.0UF 10V 10% 0402	C1, C7, C8, C11	4	Murata	GRM155Z71A105KE01J	Volume Production	Mouser	81-GRM155Z71A105KE1J	0,034	0,34
	GRM155R71C104KA88D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.1 uF 16 VDC 10% 0402 X7R	C2, C4, C5, C6, C9, C12, C13	7	Murata	GRM155R71C104KA88D	Volume Production	Mouser	81-GRM155R71C104KA88	0,018	0,18
	ZRB15XR61A475KE01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 4.7 uF 10 VDC 10% 0402 X5R	C3, C10	2	Murata	ZRB15XR61A475KE01D	Volume Production				
	LMK105ABJ474KV8F	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.47uF 10V X5R 10% SMD 0402 85°C Paper T/R	C14	1	Taiyo Yuden	LMK105ABJ474KV8F	Unknown	Mouser	963-LMK105ABJ474KV8F	0,11	0,11
	GJM1555C1H3R3WB01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 3.3 pF 50 VDC 0.05 pF 0402 C0G (NP0)	C15, C16	2	Murata	GJM1555C1H3R3WB01D	Volume Production	Mouser	81-GJM1555C1H3R3WB1D	0,14	0,28
	BLM15AG121SH1D	Ferrite Beads 120 OHM Alternate Sizing Guide Below	FB1, FB2, FB3	3	Murata	BLM15AG121SH1D	Volume Production	Digikey	490-12578-1-ND	0,1	0,3
	SMD castellated edge (Right)	9 POS Castellated connector with VIA 0.6	J1	1							
	SMD castellated edge (LEFT)	9 POS Castellated connector with VIA 0.6	J2	1							
	SMD castellated edge (Top)	9 POS Castellated connector with VIA 0.6	J3	1							
	SMD castellated edge (Bottom)	9 POS Castellated connector with VIA 0.6	J4	1							
	MLZ1608M150WT000	Inductor RF Shielded 15µH, ±5%	L1	1	TDK	MLZ1608M150WT000	Volume Production	Mouser	810-MLZ1608M150WT000	0,1	0,1
	LQW18AN8N2C80D	Inductor Wirewound , 8.2 nH, 52 Milliohm, ± 0.2nH, 4.75 GHz, 1.6 A, 0603 [1608 Metric]	L2	1	Murata	LQW18AN8N2C80D	Volume Production	Digikey	490-12792-1-ND	0,14	0,14
	CRCW040210K0JNEDC	Thick Film Resistors - SMD 1/16watt 10Kohms 5% Commercial Use	R1, R2	2	Vishay Dale	CRCW040210K0JNED	Volume Production	Newark	59M6696	0,005	0,01
	RAK3172-SIP	RAK3172-SIP is a low-power long-range transceiver based on STM32WLE5 SoC in a System-in-Package form factor.	U1	1	RAK	RAK3172-SIP-8-SM-NI	Unknown				